

Assignment - 38.

$$A = \begin{bmatrix} -4 & 72 \\ -1 & 13 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -4 - \lambda & 72 \\ -1 & 13 - \lambda \end{vmatrix} = 0$$

$$(-4 - \lambda)(13 - \lambda) + 72 = 0$$

$$-52 + 4\lambda - 13\lambda + \lambda^2 + 72 = 0$$

$$\lambda^2 - 9\lambda + 20 = 0$$

$$(\lambda - 4)(\lambda - 5) = 0$$

$$\boxed{\lambda = 4} ; \boxed{\lambda = 5} \Rightarrow \text{Eigen values} //$$

$$\lambda = 4$$

$$\begin{vmatrix} -8 & 72 \\ -1 & 9 \end{vmatrix}$$

$$-8x_1 + 72x_2 = 0$$

$$8x_1 = 72x_2$$

$$x_1 = 9x_2$$

$$V_1 = \begin{bmatrix} 9 \\ 1 \end{bmatrix}$$

$$\lambda = 5$$

$$\begin{vmatrix} -9 & 72 \\ -1 & 8 \end{vmatrix} = 0$$

$$-9x_1 + 72x_2 = 0$$

$$9x_1 = 72x_2$$

$$x_1 = 8x_2$$

$$V_2 = \begin{bmatrix} 8 \\ 1 \end{bmatrix}$$

$V_1, V_2 = \text{Eigen Vectors} //$

$$27) B = \begin{bmatrix} 5 & 9 \\ -1 & -5 \end{bmatrix}$$

$$[B - \lambda I] = 0$$

$$(5 - \lambda)(-5 - \lambda) - (9) = 0$$

$$-25 - 5\lambda + 5\lambda + \lambda^2 + 9 = 0$$

$$\lambda^2 - 16 = 0$$

$$\lambda^2 = 16$$

$$\lambda = \pm 4 \Rightarrow \text{Eigen Values.}$$

$$\lambda = 4$$

$$\begin{vmatrix} 1 & 9 \\ -1 & -9 \end{vmatrix} = 0$$

$$x_1 + 9x_2 = 0$$

$$x_1 = -9x_2$$

$$v_1 = \begin{bmatrix} -9 \\ 1 \end{bmatrix}$$

$$\lambda = -4$$

$$\begin{vmatrix} 9 & 9 \\ -1 & -1 \end{vmatrix} = 0$$

$$9x_1 + 9x_2 = 0$$

$$x_1 = -\frac{9x_2}{9}$$

$$x_1 = -x_2$$

$$v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$v_1, v_2 = \text{Eigen Vectors.}$$

$$C = \begin{bmatrix} 0 & 1 \\ 25 & 0 \end{bmatrix}$$

$$|C - \lambda I| = 0$$

$$\begin{vmatrix} \lambda & 1 \\ 25 & -\lambda \end{vmatrix} = 0$$

$$\lambda^2 - 25 = 0$$

$$\lambda^2 = 25$$

$$\lambda = \pm 5 \Rightarrow \text{Eigen Values.}$$

$$\lambda = 5$$

$$\begin{vmatrix} -5 & 1 \\ 25 & -5 \end{vmatrix}$$

$$-5x_1 + x_2 = 0$$

$$5x_1 = x_2$$

$$x_1 = \frac{x_2}{5}$$

$$v_1 = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

$$v_1, v_2 = \text{Eigen Vectors.}$$

$$\lambda = -5$$

$$\begin{vmatrix} 5 & 1 \\ 25 & 5 \end{vmatrix} = 0$$

$$5x_1 + x_2 = 0$$

$$-5x_1 = +x_2$$

$$x_1 = \frac{+x_2}{-5}$$

$$v_2 = \begin{bmatrix} +1 \\ -5 \end{bmatrix}$$

$$D = \begin{bmatrix} 5 & -2 & 3 \\ 0 & 4 & 0 \\ 0 & -1 & 3 \end{bmatrix}$$

$$|D - \lambda I| = 0$$

$$\begin{vmatrix} 5-\lambda & -2 & 3 \\ 0 & 4-\lambda & 0 \\ 0 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)(4-\lambda)(3-\lambda) = 0$$

$\lambda = 5 \quad \lambda = 4 \quad \lambda = 3 \Rightarrow$ Eigen Values. $V_{1,2,3} =$ Eigen Vectors.

$$\lambda = 5$$

$$\begin{vmatrix} 0 & -2 & 3 \\ 0 & -1 & 0 \\ 0 & -1 & -2 \end{vmatrix} = 0$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{vmatrix} 0 & -2 & 3 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{vmatrix}$$

$$-x_2 = 0$$

$$x_2 = 0$$

$$-2x_2 + 3x_3 = 0$$

$$x_3 = 0$$

$$V_1 = \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda = 4$$

$$\begin{vmatrix} 1 & -2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -1 \end{vmatrix} = 0$$

$$R_2 \leftrightarrow R_3$$

$$\begin{vmatrix} 1 & -2 & 3 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{vmatrix}$$

$$-x_2 - x_3 = 0$$

$$x_2 = -x_3$$

$$x_2 = -x_3$$

$$x_1 + 5x_3 = 0$$

$$x_1 = -5x_3$$

$$V_2 = \begin{bmatrix} -5 \\ -1 \\ 1 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{vmatrix} 2 & -2 & 3 \\ 0 & -1 & 0 \\ 0 & -1 & 0 \end{vmatrix} = 0$$

$$x_2 = 0$$

$$2x_1 - 2x_2 + 3x_3 = 0$$

$$2x_1 + 3x_3 = 0$$

$$2x_1 = -3x_3$$

$$x_1 = -\frac{3}{2}x_3$$

$$V_3 = \begin{bmatrix} -3/2 \\ 0 \\ 1 \end{bmatrix}$$

$$E = \begin{pmatrix} 1 & 0 & 12 \\ 2 & -5 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

$$|E - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & 12 \\ 2 & -5-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)[(-5-\lambda)(2-\lambda)] + 12(5+\lambda) = 0$$

$$(1-\lambda)[10+3\lambda+\lambda^2] + 60+12\lambda = 0$$

$$-10+3\lambda+\lambda^2+10\lambda+3\lambda^2-\lambda^3+60+12\lambda = 0$$

$$-\lambda^3 - 2\lambda^2 + 25\lambda + 50 = 0$$

$$\lambda^3 + 2\lambda^2 - 25\lambda - 50 = 0$$

$$\lambda^2(\lambda+2) - 25(\lambda+2) = 0$$

$$(\lambda^2 - 25)(\lambda+2) = 0$$

$$\lambda \pm 5 ; \lambda = -2 \Rightarrow \text{Eigen Values.}$$

$$\lambda = 5$$

$$\begin{bmatrix} -4 & 0 & 12 \\ 2 & -10 & 0 \\ 1 & 0 & -3 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3,$$

$$R_2 \rightarrow R_2 - 2R_1,$$

$$R_3 \rightarrow R_3 + 4R_1,$$

$$\begin{bmatrix} 1 & 0 & -3 \\ 0 & -10 & 6 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow 0$$

$$x_1 - 3x_3 = 0$$

$$x_1 = 3x_3$$

$$-10x_2 + 6x_3 = 0$$

$$10x_2 = 6x_3$$

$$x_2 = \frac{3}{5}x_3$$

$$V_1 = \begin{bmatrix} 3 \\ 3/5 \\ 1 \end{bmatrix}$$

$$\lambda = -5$$

$$\begin{bmatrix} 6 & 0 & 12 \\ 2 & 0 & 0 \\ 1 & 0 & 7 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3,$$

$$R_2 \rightarrow R_2 - 2R_1,$$

$$R_3 \rightarrow R_3 - 6R_1,$$

$$\begin{bmatrix} 1 & 0 & 7 \\ 0 & 0 & -14 \\ 0 & 0 & -30 \end{bmatrix} \Rightarrow 0$$

$$-30x_3 = 0$$

$$x_3 = 0$$

$$x_1 + 7x_3 = 0$$

$$x_1 = 0$$

$$V_2 = \begin{bmatrix} 0 \\ x \\ 0 \end{bmatrix}$$

$$\lambda = -2$$

$$\begin{bmatrix} 3 & 0 & 12 \\ 2 & -3 & 0 \\ 1 & 0 & 4 \end{bmatrix} \Rightarrow 0$$

$$R_1 \rightarrow R_1/3,$$

$$R_2 \rightarrow R_2 - 2R_1,$$

$$R_3 \rightarrow R_3 - R_1,$$

$$\begin{bmatrix} 1 & 0 & 4 \\ 0 & -3 & -8 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow 0$$

$$x_1 + 4x_3 = 0$$

$$x_1 = -4x_3$$

$$-3x_2 - 8x_3 = 0$$

$$-3x_2 = 8x_3$$

$$x_2 = -\frac{8}{3}x_3$$

$$V_3 = \begin{bmatrix} -4 \\ -8/3 \\ 1 \end{bmatrix}$$

$V_1, V_2, V_3 =$ Eigen Vectors.

$$6) F = \begin{bmatrix} -1 & 18 & 0 \\ 1 & 2 & 0 \\ 5 & -3 & -1 \end{bmatrix}$$

$$F - dI = \begin{bmatrix} -1-d & 18 & 0 \\ 1 & 2-d & 0 \\ 5 & -3 & -1-d \end{bmatrix}$$

$$(F - dI) \Rightarrow (-1-d) [(2-d)(-1-d)] - 18(-1-d) = 0$$

$$\Rightarrow (-1-d) [-2 - 2d + d + d^2 - 18] = 0$$

$$\Rightarrow (-1-d) [d^2 + 4d - 5d - 20] = 0$$

$$\Rightarrow (-1-d) [d(d+4) - 5(d+4)] = 0$$

$$\Rightarrow (-1-d)(d+4)(d-5)$$

$$\boxed{d = -1}$$

$$\boxed{d = -4}$$

$$\boxed{d = 5}$$

$\Rightarrow$ Eigen Values: //

$$d = 1$$

$$\begin{bmatrix} \cancel{0} & 18 & 0 \\ 1 & 3 & 0 \\ 5 & -3 & 0 \end{bmatrix}$$

$$R_2 \leftrightarrow R_1$$

$$R_3 \rightarrow R_3 - 5R_1$$

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 18 & 0 \\ 0 & -18 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 + 3x_2 = 0$$

$$x_1 = -3x_2$$

$$18x_2 = 0$$

$$V_1 = \begin{bmatrix} 0 \\ 0 \\ x \end{bmatrix}$$

$n = 3$

$$\begin{pmatrix} 3 & 18 & 0 \\ 1 & 6 & 0 \\ 5 & -3 & 3 \end{pmatrix}$$

$$\begin{aligned} R_2 &\rightarrow 3R_2 - R_1 \\ R_3 &\rightarrow 3R_3 - 5R_1 \end{aligned}$$

$$\begin{pmatrix} 3 & 18 & 0 \\ 0 & 0 & 0 \\ 0 & -99 & 9 \end{pmatrix}$$

$$3x_1 + 18x_2 = 0$$

$$x_1 = -6x_2$$

$$-99x_2 + 9x_3 = 0$$

$$9x_3 = 99x_2$$

$$x_3 = 11x_2$$

$$V_2 = \begin{bmatrix} -6 \\ 1 \\ 11 \end{bmatrix}$$

$V_1, V_2, V_3 = \text{Eigen Vektors.}$

$n = 5$

$$\begin{pmatrix} -6 & 18 & 0 \\ 1 & -3 & 0 \\ 5 & -3 & -6 \end{pmatrix}$$

$$\begin{aligned} R_2 &\rightarrow 6R_2 + R_1 \\ R_3 &\rightarrow 6R_3 + 5R_1 \end{aligned}$$

$$\begin{pmatrix} -6 & 18 & 0 \\ 0 & 0 & 0 \\ 0 & 72 & -36 \end{pmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$\begin{pmatrix} -6 & 18 & 0 \\ 0 & 72 & -36 \\ 0 & 0 & 0 \end{pmatrix}$$

$$-6x_1 + 18x_2 = 0$$

$$6x_1 = 18x_2$$

$$x_1 = 3x_2$$

$$72x_2 - 36x_3 = 0$$

$$36x_3 = 72x_2$$

$$x_3 = 2x_2$$

$$V_3 = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$